



Universidade do Minho
Escola de Engenharia
Departamento de Informática

Introdução e Instalação do Knime

MESTRADO EM ENGENHARIA DE TELECOMUNICAÇÕES E INFORMÁTICA
Inteligência Artificial para as Telecomunicações
2025/2026



KNIME Analytics Platform

- ❖ **Open source** platform used to **automate data science process**;
- ❖ Provides tools to **access, transform and clean data, train algorithms, perform deep learning, create interactive visualizations, ...**

Core Features:

- ❖ **Read and merge data** from **different file formats**: CSV, XML, JSON, PDF, ...
- ❖ Create **Machine Learning models** used for **grouping, regression or classification**.

Application tabs

Entry page tab and all opened workflows tabs.

Side panel navigation

Description

Description of the displayed workflow or component.

Node repository

All available nodes in KNIME Analytics Platform to build your workflows.

Space explorer

Navigate local or KNIME Hub spaces and access your workflows, components and files.

Workflow Editor

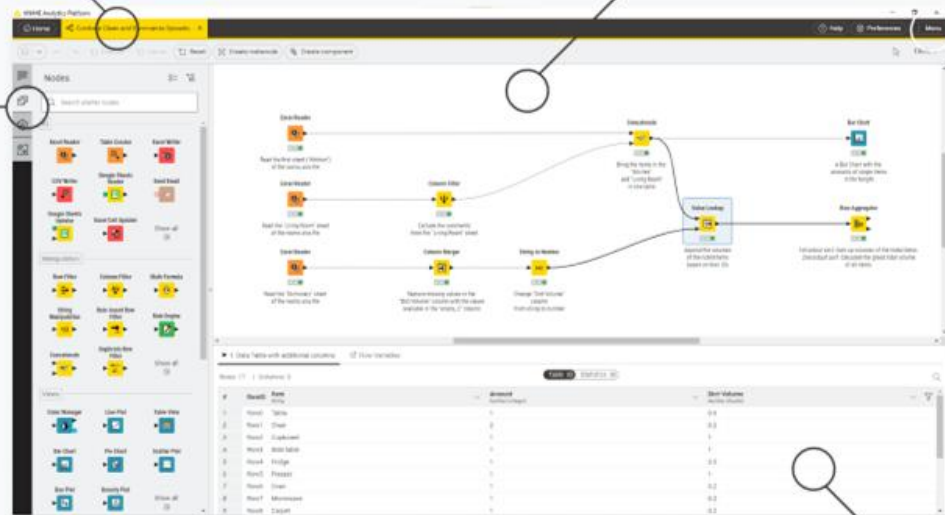
Canvas for editing the currently active workflow.

Help, Preferences, Menu

Access to more material, install additional extensions and change settings for the node repository.

Node Monitor

Shows the output of the current selected node and also the flow variable values.



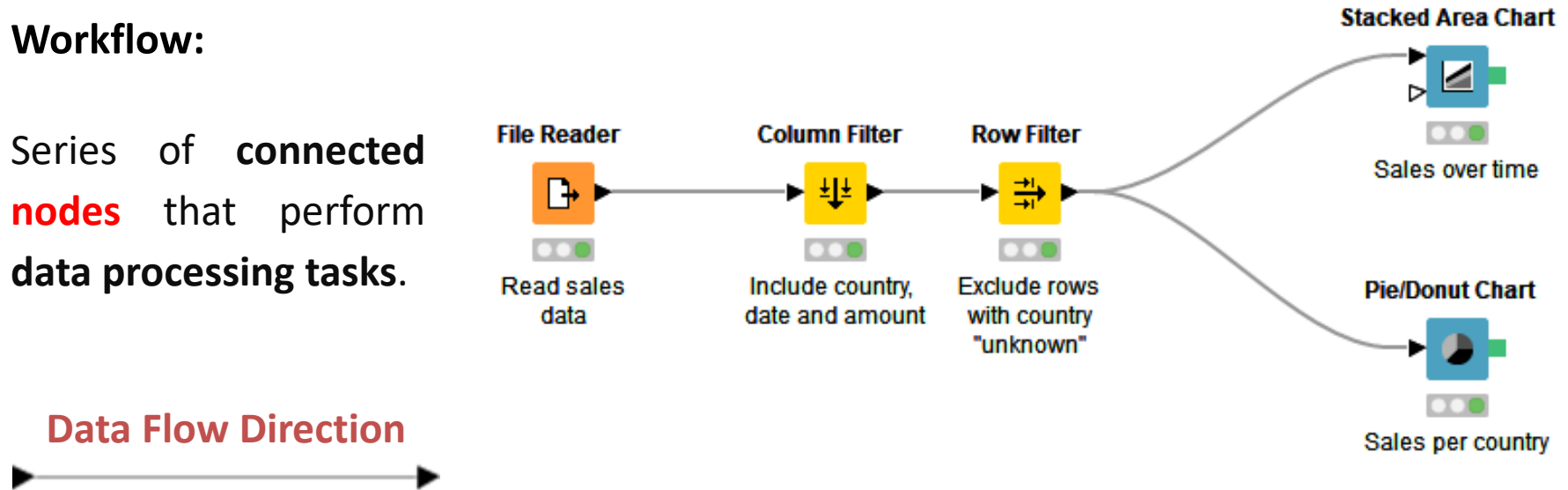
KNIME - Workflow Project Example



KNIME: Core Elements

Workflow:

Series of **connected nodes** that perform **data processing tasks**.



KNIME: Core Elements

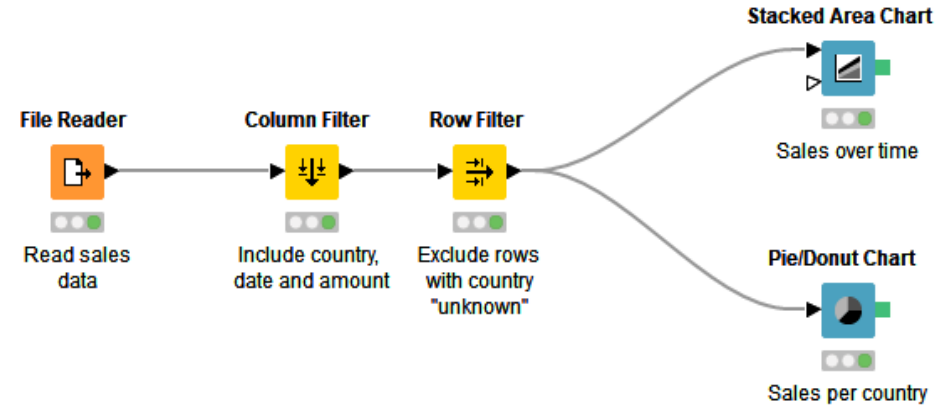
File Reader: Read sales data from a file;

Column Filter: Only include the data related to country, date and amount;

Row Filter: Remove unknown values related to the country;

Stacked Area Chart + Pie Chart:

Visualize the evolution of the amount of sales.

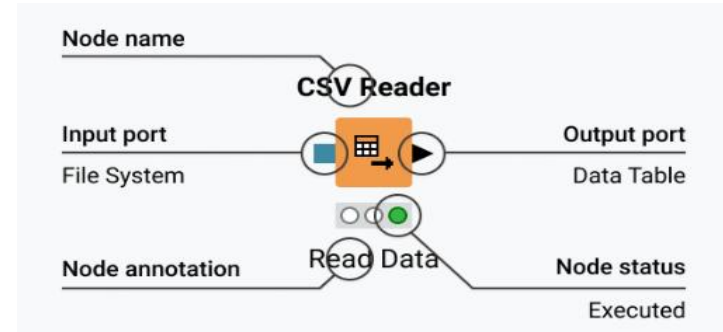


KNIME: Core Elements

Nodes:

Individual components or functions used to **perform specific tasks**.

Building blocks of the data analysis or processing pipelines.



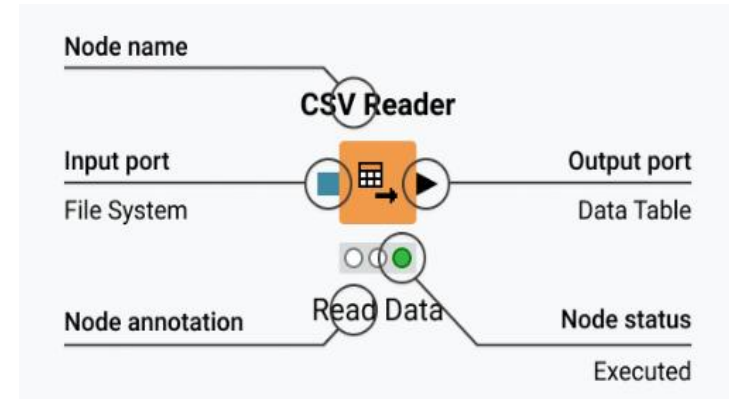
KNIME: Core Elements

Nodes: Input and Output Ports

Input ports: Receive data from other nodes;

Output ports: Send data to other nodes.

The output port of a node connects to the input port of the following node.



KNIME: Core Elements

Nodes: Status

Nodes can have **4 different status**



Not configured

The node is waiting for configuration or incoming data.



Configured

The node has been configured correctly, and can be executed.



Executed

The node has been successfully executed. Results may be viewed and used in downstream nodes.



Error

The node has encountered an error during execution.

KNIME: Core Elements

Nodes: Action Bar

Configure: The configure button most to the left in the node action bar **opens the configuration panel** of the node.

Execute: The play button in the node action bar **executes the node**.

Cancel: The x button in the node action bar **stops the execution** of the node.

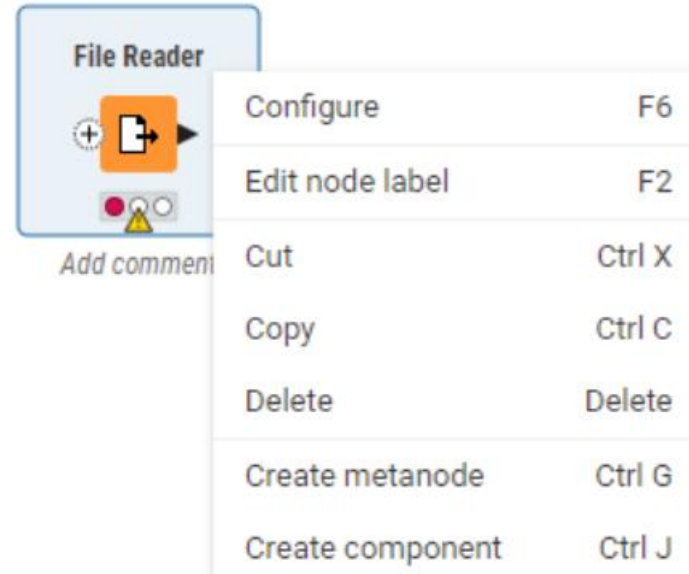
Reset: The button showing an arrow that points to the left **resets the node**.



Nodes: Open Configuration Panel

3 different ways:

- In the node action bar, click the **settings icon**;
- **Right click** the node + **Configure**;
- **Double click** the node + **Configure**.



Example: File Reader node's configuration panel

File

Settings Transformation Advanced Settings Limit Rows Encoding Flow Variables Job Manager Selection Memory Policy

Input location

Read from Local File System

Mode ☒ File ☐ Files in folder

File Browse... 

 Please specify a file

Reader options

Format

Autodetect format 

Column delimiter Row delimiter ☒ Line break ☐ Custom

Quote char Quote escape char

Comment char

☒ Has column header ☐ Has row ID

☐ Support short data rows ☐ Prepend file index to row ID

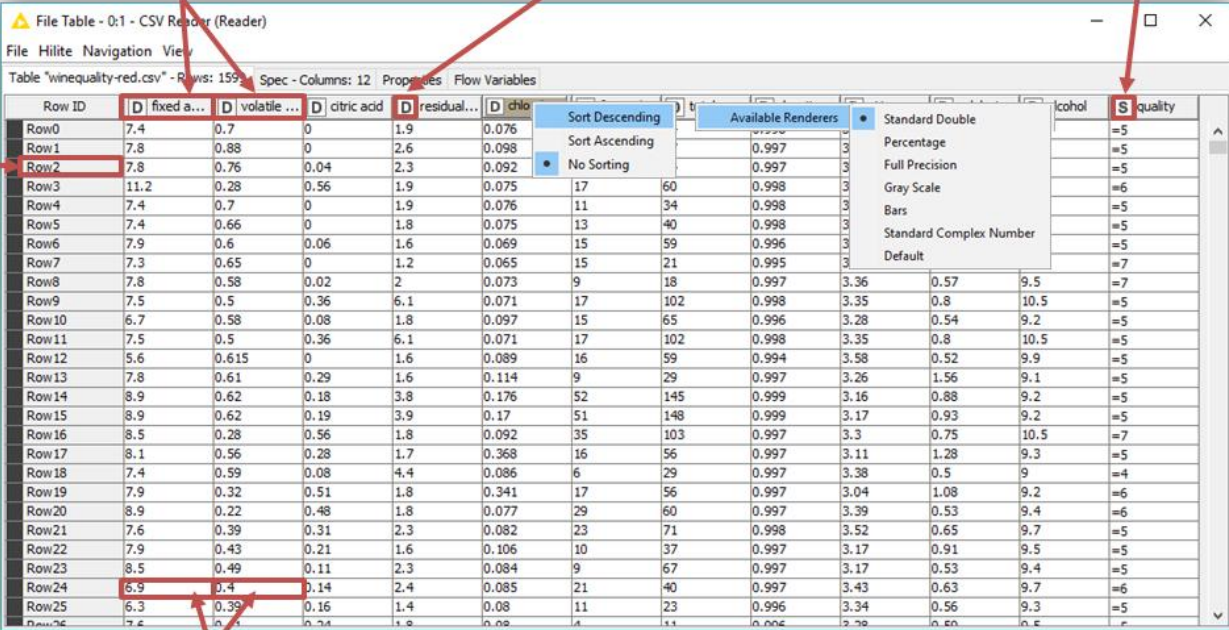
Column Headers

Data Type (Double)

Data Type (String)

Row ID

Data Cells



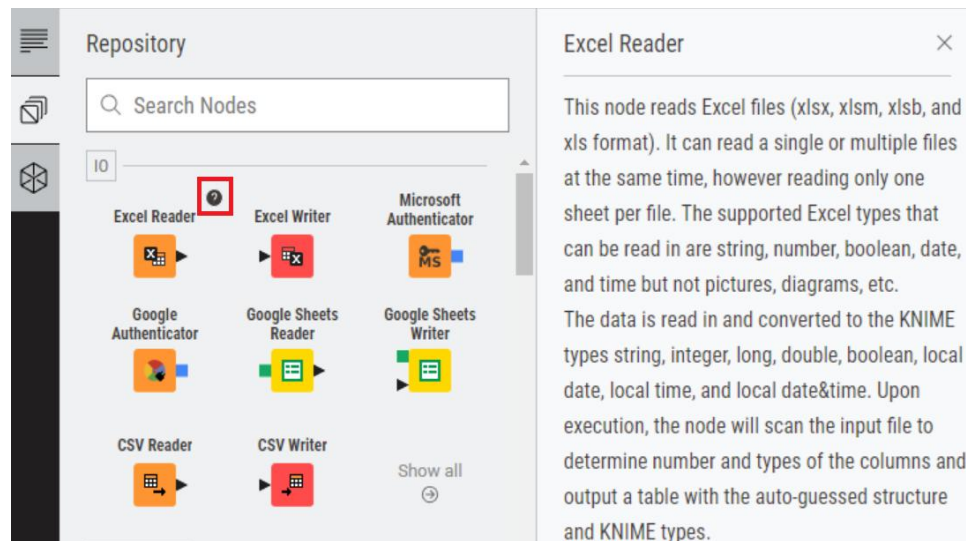
Row ID	fixed a...	volatile ...	citric acid	residual...	chlo	alcohol	quality
Row0	7.4	0.7	0	1.9	0.076		=5
Row1	7.8	0.88	0	2.6	0.098		=5
Row2	7.8	0.76	0.04	2.3	0.092		=5
Row3	11.2	0.28	0.56	1.9	0.075		=6
Row4	7.4	0.7	0	1.9	0.076		=5
Row5	7.4	0.66	0	1.8	0.075		=5
Row6	7.9	0.6	0.06	1.6	0.069		=5
Row7	7.3	0.65	0	1.2	0.065		=7
Row8	7.8	0.58	0.02	2	0.073	3.36	=7
Row9	7.5	0.5	0.36	6.1	0.071	3.35	=5
Row10	6.7	0.58	0.08	1.8	0.097	3.28	=5
Row11	7.5	0.5	0.36	6.1	0.071	3.35	=5
Row12	5.6	0.615	0	1.6	0.089	3.58	=5
Row13	7.8	0.61	0.29	1.6	0.114	3.26	=5
Row14	8.9	0.62	0.18	3.8	0.176	3.16	=5
Row15	8.9	0.62	0.19	3.9	0.17	3.17	=5
Row16	8.5	0.28	0.56	1.8	0.092	3.3	=7
Row17	8.1	0.56	0.28	1.7	0.368	3.11	=5
Row18	7.4	0.59	0.08	4.4	0.086	3.38	=4
Row19	7.9	0.32	0.51	1.8	0.341	3.04	=6
Row20	8.9	0.22	0.48	1.8	0.077	3.39	=6
Row21	7.6	0.39	0.31	2.3	0.082	3.52	=5
Row22	7.9	0.43	0.21	1.6	0.106	3.17	=5
Row23	8.5	0.49	0.11	2.3	0.084	3.17	=5
Row24	6.9	0.4	0.14	2.4	0.085	3.43	=6
Row25	6.3	0.39	0.16	1.4	0.08	3.34	=5

Example: After inserting a path to a CSV file

Nodes: How to know what a node does?

In the **Node Repository**:

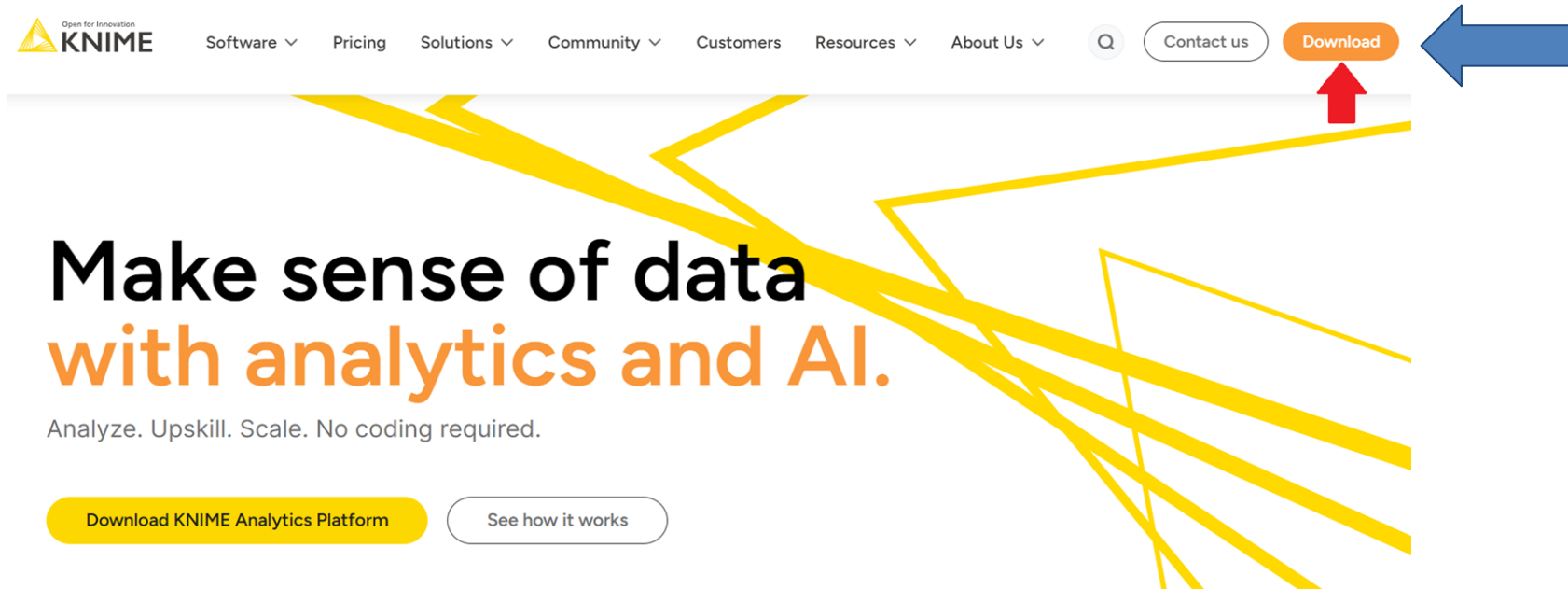
- Click on the **question mark symbol**, on the top right corner of the desired node.



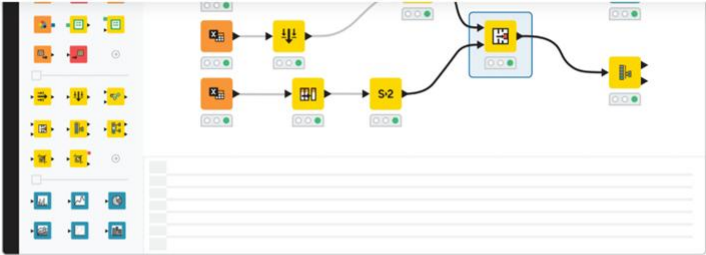
How to install KNIME Analytics Platform?

- ❖ Go to **KNIME's website**: <https://www.knime.com/>
- ❖ Click the **Download** button
- ❖ **Accept the** terms and conditions and click Download
- ❖ Select your **operating system** (*Windows*) and download the **Installer**

How to install KNIME Analytics Platform?

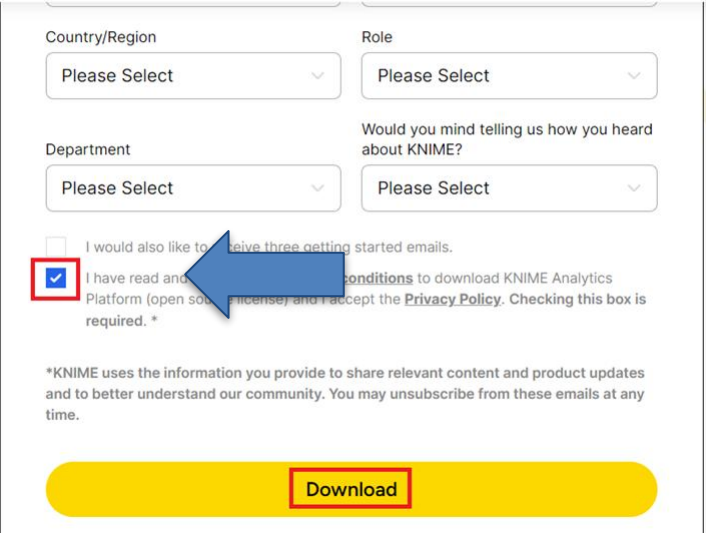


How to install KNIME Analytics Platform?



KNIME Analytics Platform is a free and open source low-code/no-code software that allows anyone, irrespective of experience levels, to make sense of data.

Download today, and use it free forever.



Country/Region
Please Select

Role
Please Select

Department
Please Select

Would you mind telling us how you heard about KNIME?
Please Select

☐ I would also like to receive three getting started emails.

☒ I have read and agree to the [Terms and Conditions](#) to download KNIME Analytics Platform (open source license) and I accept the [Privacy Policy](#). Checking this box is required. *

*KNIME uses the information you provide to share relevant content and product updates and to better understand our community. You may unsubscribe from these emails at any time.

Download

How to install KNIME Analytics Platform?

Windows

Microsoft Defender SmartScreen may block download in its attempt to prevent malicious software installations.

To solve the problem [click here](#).

KNIME Analytics Platform for Windows (installer)

[Download](#) (570MB)

KNIME Analytics Platform for Windows (self-extracting archive)

[Download](#) (582MB)

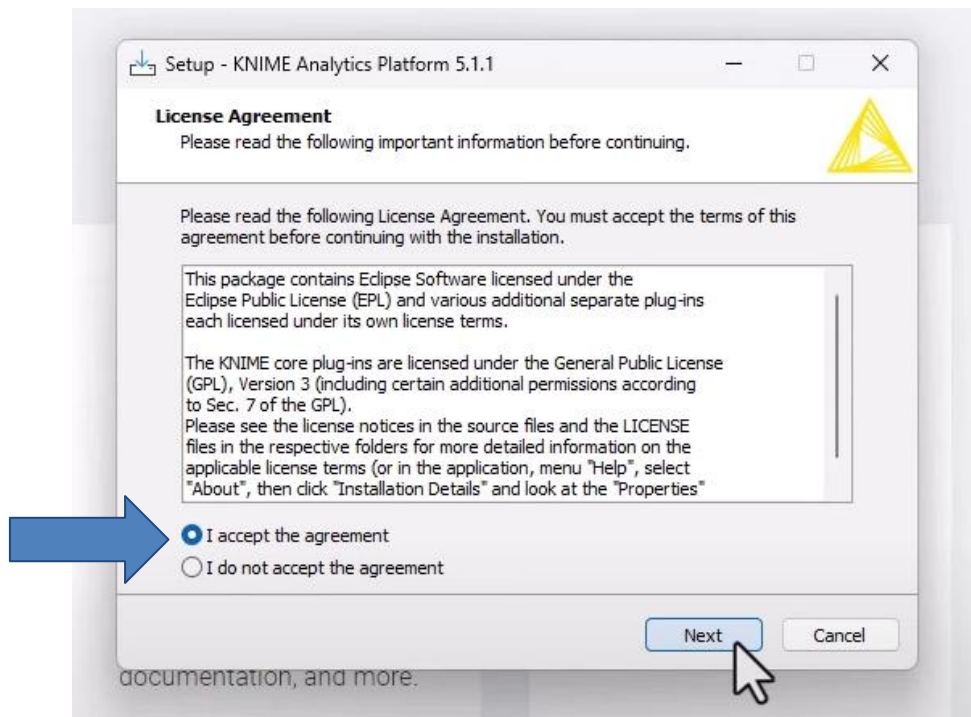
KNIME Analytics Platform for Windows (zip archive)

[Download](#) (692MB)

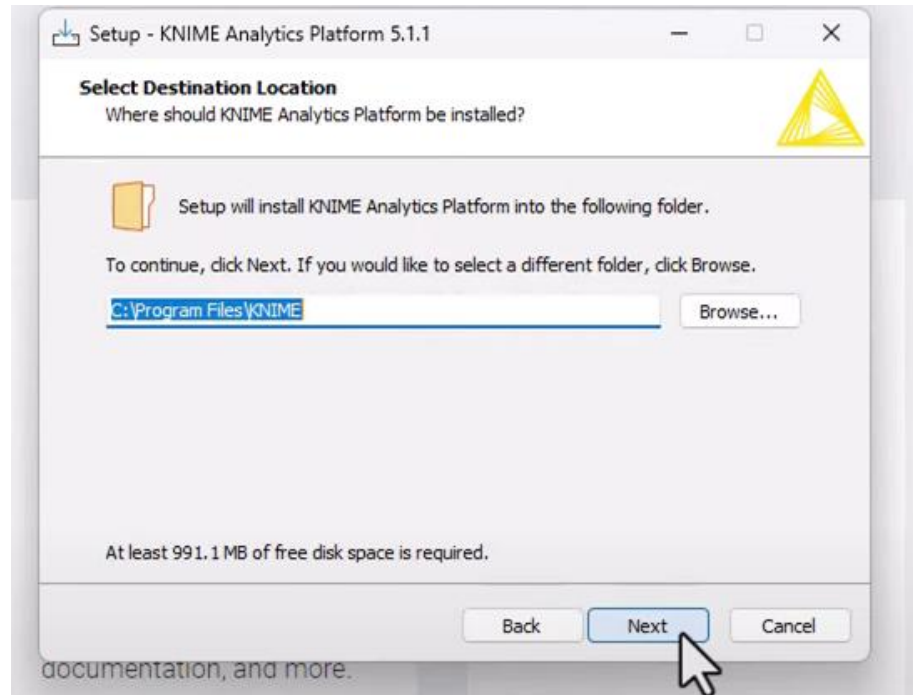
How to install KNIME Analytics Platform?

- ❖ Open the **installer file**
- ❖ Accept the **License Agreement** and click **Next**
- ❖ Select the desired **installation folder** and click **Next**

How to install KNIME Analytics Platform?

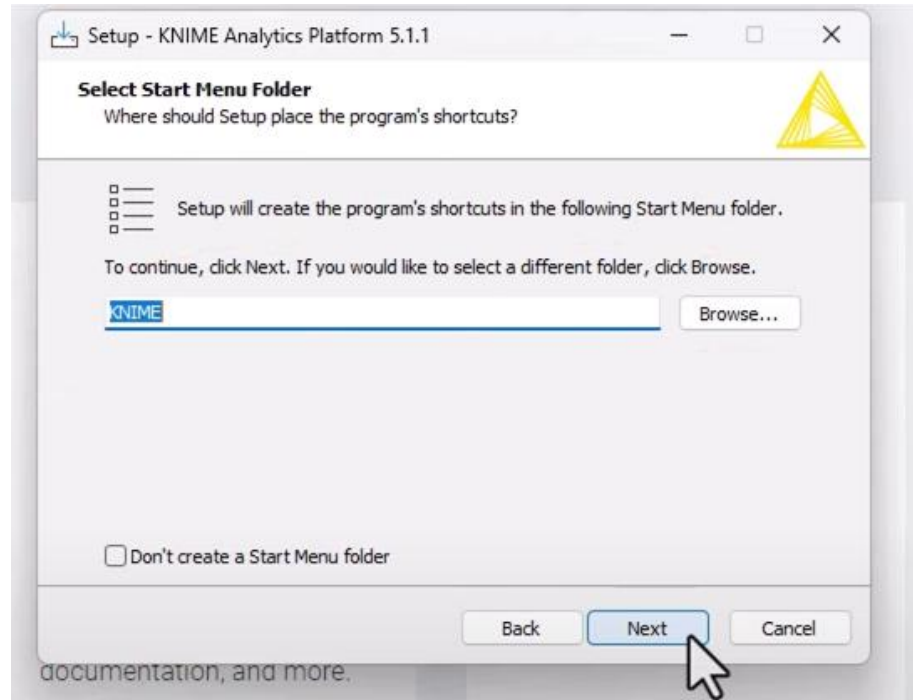


How to install KNIME Analytics Platform?



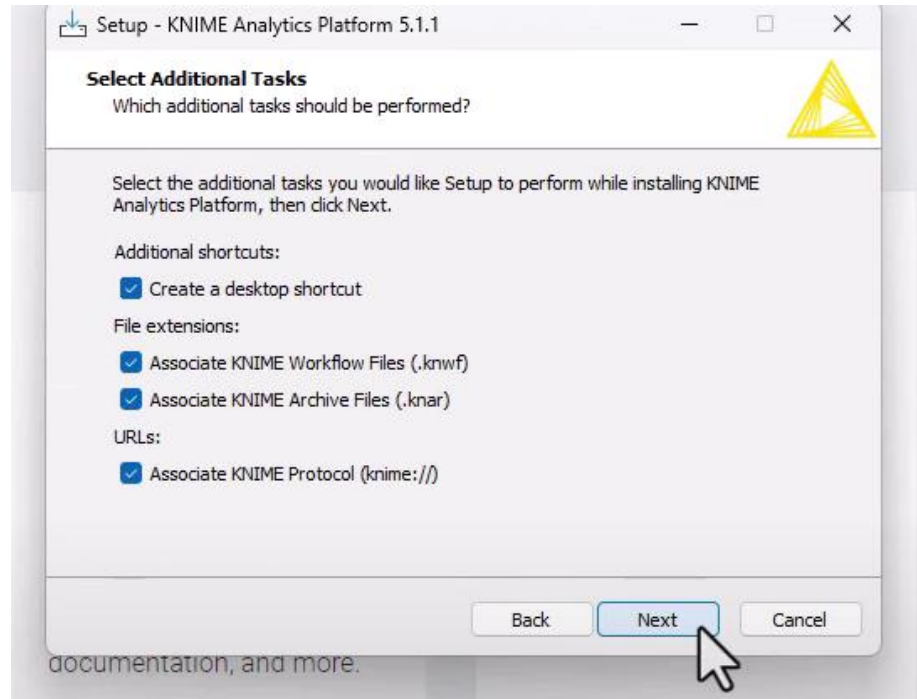
How to install KNIME Analytics Platform?

Click **Next** on the following dialog windows



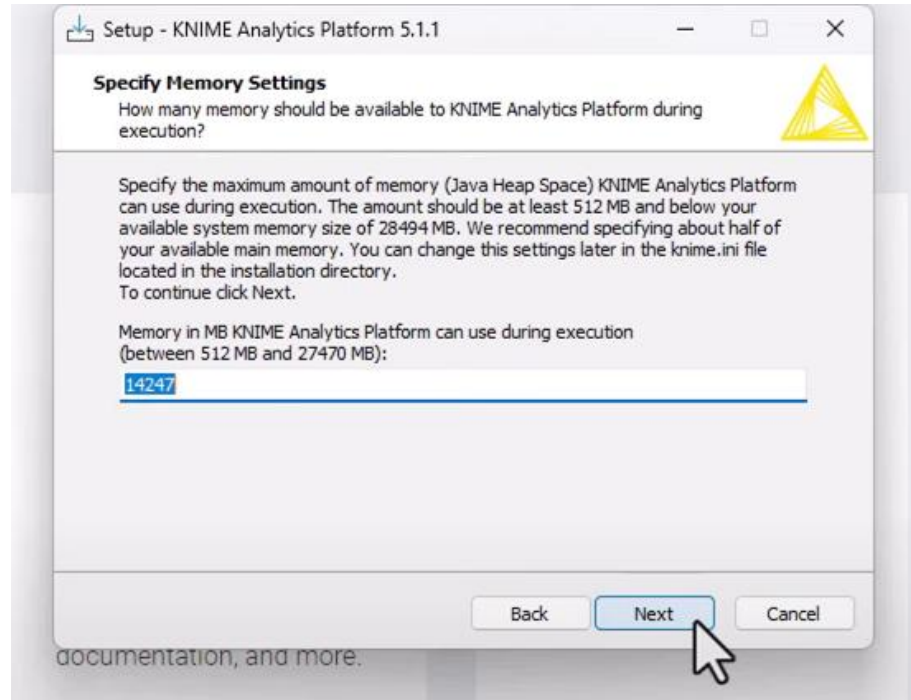
How to install KNIME Analytics Platform?

Click **Next** on the following dialog windows



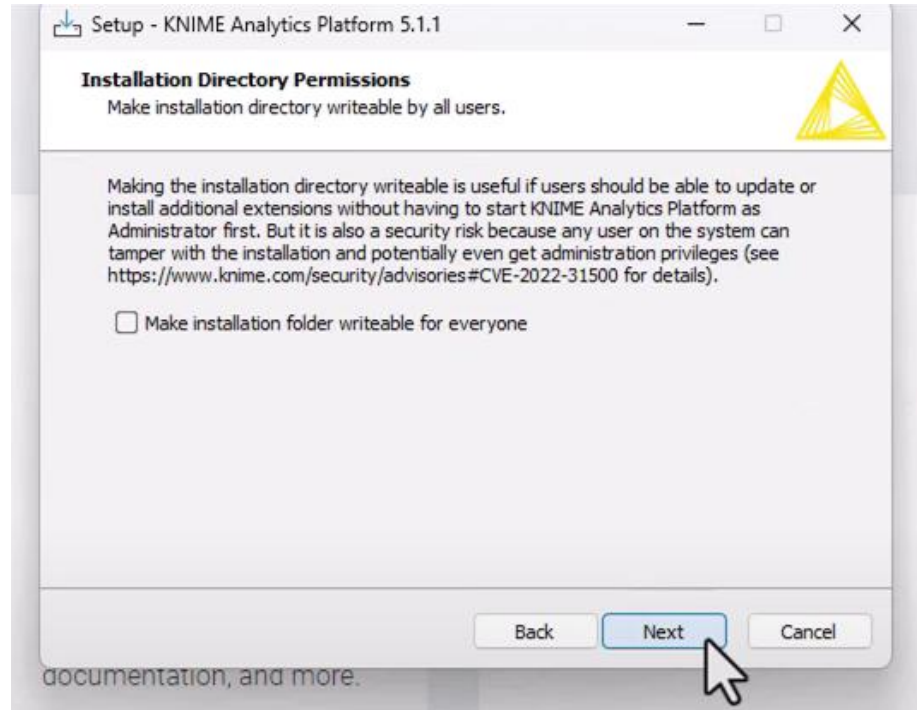
How to install KNIME Analytics Platform?

Click **Next** on the following dialog windows



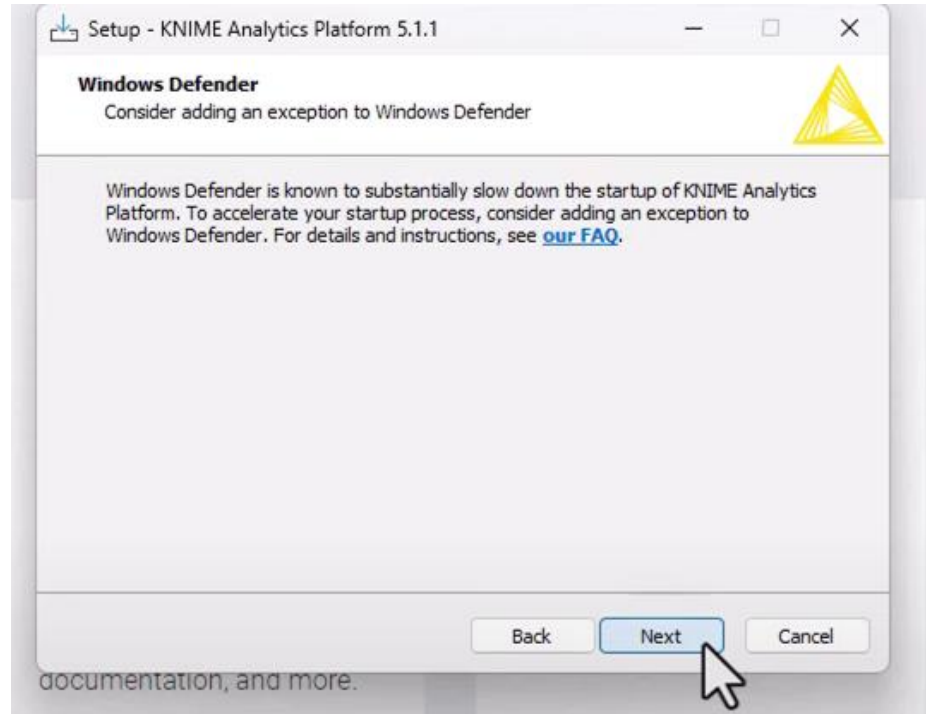
How to install KNIME Analytics Platform?

Click **Next** on the following dialog windows



How to install KNIME Analytics Platform?

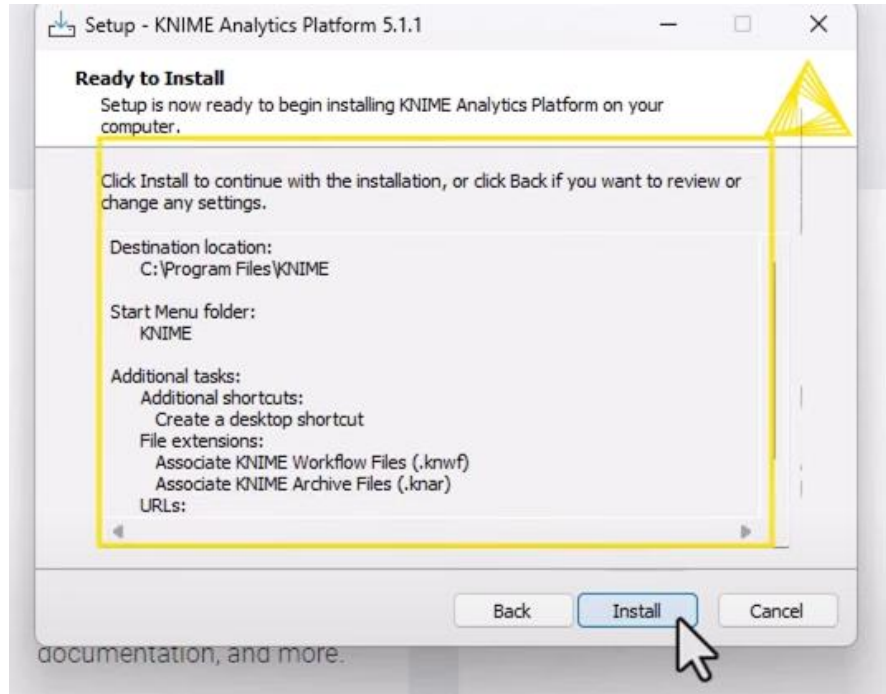
Click **Next** on the following dialog windows



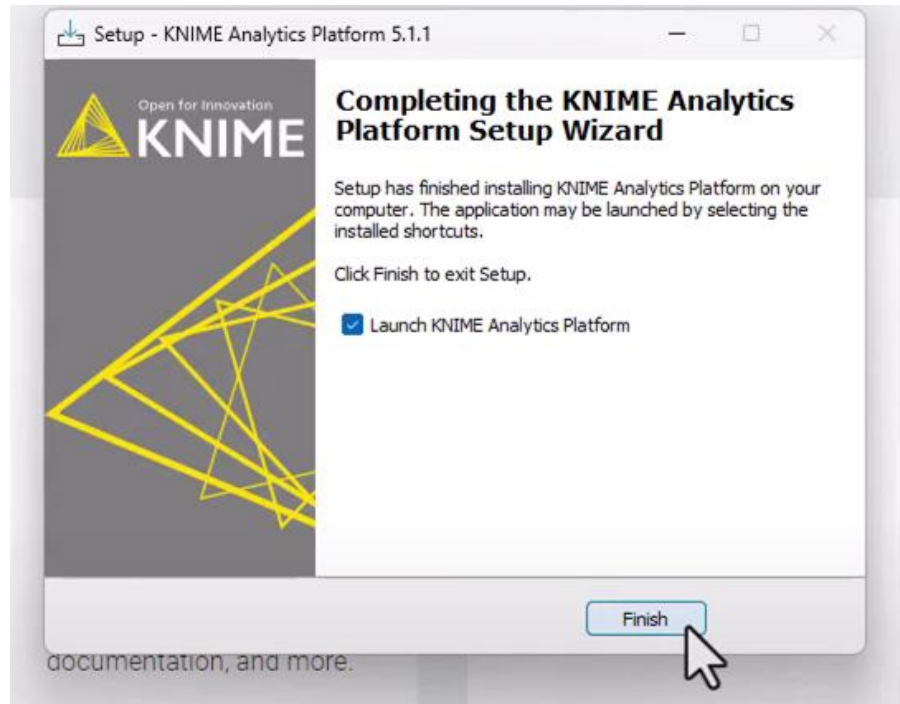
How to install KNIME Analytics Platform?

- ❖ You should reach a window with a **summary of the installation settings**
- ❖ Click **Install**
- ❖ Once the settings are installed, click **Finish** to launch the program

How to install KNIME Analytics Platform?



How to install KNIME Analytics Platform?





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